

UME745 : Machine Tool Design

L	T	P	Cr
3	0	0	3.0

Course Objective: The objective of this course is to develop the basic skills and understanding on the working principles, mechanics, technological capabilities, design philosophy of machine tool elements and their subsystems.

Syllabus

Introduction: Classification of machine tools, General requirements of machine tool design, Engineering design process applied to machine tools.

Machine tool drives: Mechanical, hydraulic and electrical drives, speed and feed regulations, design of speed box and feed box, machine tool control systems.

Design of machine tool structures: Basic design procedure of machine tool structures for strength & stiffness, dynamics of machine tools, design of bed, head stock, housing, etc., design of spindles and spindle supports, design of hydrostatic, hydrodynamic and antifriction guideways, design calculations for lead screw and ball re-circulating power screw assemblies.

Design considerations in CNC machine tools: Special features, constructional details and design considerations in CNC machines.

Research Assignment

Students in the group will submit a research assignment or design project based on the design and analysis of machine tool/components.

Design assignment will include literature review on the recent technology developments, identification of the operational requirements and industrial applications of the selected machine tool, selection of drive system and control system, designing of various structural components.

Design project may include refining the existing design of the selected machine, preparation of questionnaire and feedbacks, geometric modeling, engineering analysis and optimization of modeled structural components and generating engineering drawings of the complete machine or subsystems of a selected machine. Every group will be required to present their works and submit a final technical report at the end of the semester.

Course Learning Objectives (CLO)

The students will be able to:

1. Analyze the design philosophy and design process adopted for the development of machine tools.
2. Analyze the constructions and structural behavior of a machine tool.
3. Analyze the drive and control systems used in machine tools

4. Design the components and subsystems of a given machine.

Text Books

1. Mehta, N. K., Machine Tool Design & Numerical Control, McGraw Hill (2012), 3rd ed.
2. Shigley, J., Mechanical Engineering Design, McGraw Hill, New York (2014), 10th ed.
3. Rao P.N., Manufacturing Technology: Metal cutting and Machine Tools, McGraw Hill (2013), 3rd ed.

Reference Books

1. Basu, S. K. and Pal, D.K., Design of Machine Tools, Allied Publishers (2008), 5th ed.
2. Acherkhan, N.S., Machine Tool Design, University Press of the Pacific, (2000).
3. Boothroyd, G. and Knight Wiston, A., Fundamentals of Machining And Machine Tools, CRC Press (2005), 3rd Edition.
4. Sharma, P. C., A Text Book Of Machine Tools & Tool Design, S. Chand Limited, (2005)

Evaluation Scheme

S. No.	Evaluation Elements	Weightage (%)
1 .	MST	30
2 .	EST	45
3 .	Sessionals (Assignments/Tutorials/Quizzes)	25

UME741 : Metal Forming